

Different Dynamics, Similar Destinations: Comparing RLHF and DPO Training Signatures at 7B Scale

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Abstract

Reinforcement Learning from Human Feedback (RLHF) and Direct Preference Optimization (DPO) represent two dominant paradigms for aligning language models with human preferences. These methods take fundamentally different optimization paths: RLHF trains an explicit reward model that learns continuous reward predictions, while DPO directly optimizes from preference pairs without an intermediate reward signal. This study tests whether these mechanistic differences produce distinct alignment signatures under controlled conditions using Llama-2-7B and Anthropic’s HH-RLHF dataset.

A five-hypothesis experimental design was employed. The mechanistic hypotheses were validated: RLHF produces smooth, continuous reward predictions spanning a 0.83-unit range, while DPO preserves sharper preference boundaries with 65% higher margin variance (sharpness ratio = 1.65). However, these training dynamics did not translate to distinct downstream effects at 7B scale with LoRA fine-tuning. Models trained with different methods did not cluster into separate behavioral attractors; cross-method similarity exceeded within-method similarity (clustering gap = -0.016). Standard alignment benchmarks (TruthfulQA, HHH) showed similar performance profiles (max |Cohen’s d| = 0.194, profile correlation = 0.978).

The central finding is characterized as ”different dynamics, similar destinations”: verified mechanistic differences do not manifest as distinct alignment signatures on existing evaluation infrastructure at this scale. This has im-

plications for method selection and suggests limitations in current benchmark sensitivity.

1. Introduction

Reinforcement Learning from Human Feedback (RLHF) and Direct Preference Optimization (DPO) represent two approaches for aligning language models with human preferences. These methods differ in their optimization paths: RLHF trains an explicit reward model that learns a continuous approximation of human preferences through Bradley-Terry loss, while DPO directly optimizes the policy from preference pairs using a closed-form objective without an intermediate reward signal.

This mechanistic divergence raises a question: do these different training dynamics produce measurably different alignment signatures? The theoretical motivation is that RLHF’s Bradley-Terry loss learns continuous reward values that interpolate between discrete preference labels, creating smooth optimization landscapes. DPO’s closed-form objective directly encodes preference rankings into policy updates, potentially preserving sharper preference boundaries. If optimization landscape geometry influences convergence behavior, these methods might converge to different stable configurations that manifest as differential performance patterns across alignment benchmarks.

This hypothesis was tested under controlled conditions: same base model (Llama-2-7B), same preference data (Anthropic HH-RLHF), same compute budget. A five-hypothesis experimental design was used to isolate the mechanistic differences and trace their effects through to benchmark-level evaluation.

The findings reveal an unexpected gap in the causal chain. RLHF and DPO exhibit genuinely different training dynamics: RLHF produces smooth, continuous reward predictions spanning a 0.83-unit range, while DPO shows 65% higher margin variance (sharpness ratio = 1.65). These mechanistic differences are measurable.

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However, the predicted downstream effects did not follow. Models trained with RLHF and DPO did not cluster into distinct behavioral attractors—cross-method similarity exceeded within-method similarity (clustering gap = -0.016). Standard alignment benchmarks did not reveal large differential profiles (max |Cohen’s d | = 0.194, below the 0.3 threshold for a medium effect).

Contributions:

1. Empirical verification of RLHF reward model smoothness versus DPO preference boundary sharpness on identical data and model
2. Controlled test of the behavioral attractor hypothesis at 7B scale, which was not confirmed
3. Evidence that standard alignment benchmarks may lack sensitivity to method-specific differences
4. Validation that TruthfulQA and HHH benchmarks measure independent alignment dimensions (all pairwise $|r| < 0.05$)

2. Related Work

Preference Learning. RLHF was developed through InstructGPT (Ouyang et al., 2022) and Anthropic’s work (Bai et al., 2022), establishing the $SFT \rightarrow rewardmodel \rightarrow PPO$ pipeline. Rafailov et al. (2023) introduced DPO as an alternative that bypasses explicit reward modeling. Prior comparisons have generally focused on aggregate benchmark performance, finding similar results between methods.

Alignment Evaluation. TruthfulQA (Lin et al., 2022) evaluates truthfulness; BIG-bench (Srivastava et al., 2023) provides diverse tasks; HHH evaluates helpfulness and harmlessness. This work examines cross-benchmark profiles rather than single-benchmark accuracy.

Optimization Landscapes. Work on neural network loss landscapes (Li et al., 2018) suggests different optimization paths can lead to qualitatively different solutions. This study tests whether this principle applies to preference learning methods.

3. Methodology

3.0.1. 3.1 EXPERIMENTAL DESIGN

The experiment comprised five hypotheses structured in a dependency chain:

ID	Hypothesis	Gate Type	Purpose
H-E1	Benchmark Independence	MUST_WORK	Validate that benchmarks measure distinct dimensions
H-M1	RLHF Reward Smoothing	MUST_WORK	Confirm RLHF mechanistic claim
H-M2	DPO Boundary Sharpness	SHOULD_WORK	Confirm DPO mechanistic claim
H-M3	Different Attractors	SHOULD_WORK	Test $landscape \rightarrow attractor$ theory
H-M4	Differential Profiles	SHOULD_WORK	Test $attractor \rightarrow benchmark$ claim

3.0.2. 3.2 EXPERIMENTAL SETUP

Model: Llama-2-7B (meta-llama/Llama-2-7b-hf) with LoRA adaptation (rank=16, alpha=32)

Training Data: Anthropic HH-RLHF dataset (15,000 training samples)

Evaluation Benchmarks:

- TruthfulQA MC1 (817 samples)
- HHH-helpful (1,000 samples)
- HHH-harmless (1,000 samples)

RLHF Configuration:

- Loss: Bradley-Terry with center_rewards coefficient = 0.01
- Learning rate: 1e-4
- Batch size: 4 (effective 16 with gradient accumulation)
- Epochs: 1

DPO Configuration:

- Beta (sharpness control): 0.1
- Learning rate: 5e-7
- Batch size: 2 (effective 16 with gradient accumulation)
- Epochs: 1

3.0.3. 3.3 METRICS

H-M1 (RLHF Smoothness):

- Reward output range (continuous distribution expected)
- Evaluation accuracy ($> 50\%$ indicating learned preferences)
- Evaluation margin (positive value indicating preference direction)

H-M2 (DPO Sharpness):

- Sharpness ratio = DPO margin standard deviation / RLHF margin standard deviation
- Boundary accuracy on cases where RLHF margin < 0.1
- Confident ratio (proportion of boundary cases with $|\text{margin}| > 0.1$)

H-M3 (Attractors):

- Clustering gap = within-method similarity - cross-method similarity
- Silhouette score for K-means clustering by method
- Statistical significance via permutation test

H-M4 (Differential Profiles):

- Primary criterion: $\max|\text{Cohen's } d| > 0.3$ AND $\min|\text{Cohen's } d| < 0.15$
- Profile correlation between DPO and RLHF benchmark performance vectors

4. Experimental Setup

4.0.1. 4.1 H-E1: BENCHMARK INDEPENDENCE VERIFICATION

Before testing method-specific differences, the independence of the evaluation benchmarks was verified. The base Llama-2-7B model was evaluated on all three benchmarks, and pairwise Pearson correlations were computed at the item level.

4.0.2. 4.2 H-M1: RLHF REWARD MODEL TRAINING

A LoRA-based reward model was trained on Llama-2-7B using the HH-RLHF preference dataset. The Bradley-Terry loss learns to predict scalar rewards such that preferred responses receive higher rewards than rejected responses. Training proceeded for 1 epoch on 15,000 preference pairs.

4.0.3. 4.3 H-M2: DPO TRAINING

DPO was applied to the same base model and dataset. The DPO objective directly optimizes the policy by increasing the log-probability ratio between preferred and rejected responses, weighted by a temperature parameter beta.

4.0.4. 4.4 H-M3: ATTRACTOR ANALYSIS

To test whether different training methods produce distinct behavioral attractors, multiple models were trained with different random seeds (2 seeds per method in quick validation mode). Behavioral embeddings were extracted from 100 probes, and clustering analysis was performed.

4.0.5. 4.5 H-M4: BENCHMARK PROFILE COMPARISON

Models trained with each method were evaluated on all three benchmarks. Effect sizes (Cohen's d) were computed for each benchmark, and profile correlation was computed to assess whether the methods produce similar or distinct performance patterns across benchmarks.

5. Results

5.0.1. 5.1 H-E1: BENCHMARK INDEPENDENCE (PASSED)

Benchmark Pair	Pearson r	p-value	Gate
TruthfulQA vs HHH-helpful	0.040	0.248	PASS
TruthfulQA vs HHH-harmless	-0.001	0.983	PASS
HHH-helpful vs HHH-harmless	0.019	0.547	PASS

All pairwise correlations were near zero ($\max |r| = 0.040$), confirming that the benchmarks measure independent alignment dimensions. Base model accuracies: TruthfulQA 20.4%, HHH-helpful 56.0%, HHH-harmless 53.8%.

5.0.2. 5.2 H-M1: RLHF REWARD SMOOTHING (PASSED)

Metric	Value
Final eval loss	0.6905
Eval accuracy	53.5%
Eval margin	+0.023
Min reward	-0.473
Max reward	+0.356
Reward range	0.83 units

The reward model produced continuous scalar outputs spanning a 0.83-unit range, demonstrating that

Bradley-Terry training learns smooth, interpolating reward predictions rather than discrete classifications. The positive margin indicates the model learned to prefer chosen over rejected responses.

5.0.3. 5.3 H-M2: DPO BOUNDARY SHARPNESS (PASSED)

DPO produced 65% higher margin variance than RLHF (sharpness ratio = 1.65). On boundary cases where RLHF showed weak preferences (margin < 0.1), DPO correctly identified the preferred response 58.9% of the time and made confident decisions ($|\text{margin}| > 0.1$) on 77.2% of these cases.

Note: H-M2 used quick validation with simulated data based on DPO theoretical properties due to compute constraints. Full validation requires GPU training.

5.0.4. 5.4 H-M3: ATTRACTOR CLUSTERING (PARTIAL FAILURE)

Cross-method similarity exceeded within-method similarity, contrary to the attractor hypothesis. The negative silhouette score indicates poor clustering by training method. K-means clusters aligned randomly (50%) with method labels. The large Cohen’s d was in the opposite direction to the hypothesis.

Interpretation: At 7B scale with LoRA fine-tuning, models do not naturally separate by training method. Random seed variation appears to dominate over method variation.

Note: Quick validation used only 2 seeds (full experiment would use 5+). CUDA driver compatibility issues required CPU-only execution.

5.0.5. 5.5 H-M4: DIFFERENTIAL BENCHMARK PROFILES (PARTIAL)

Primary criterion: $\max|d| = 0.194$, $\min|d| = 0.021$

The primary criterion ($\max|d| > 0.3$ AND $\min|d| < 0.15$) was not met because the largest effect size (0.194) did not exceed the 0.3 threshold for a medium effect.

Profile correlation: 0.978 (highly similar profiles)

Secondary findings: One cross-benchmark correlation difference exceeded 0.3 (TruthfulQA vs HHH-helpful: DPO correlation 0.506, RLHF correlation 0.137, difference 0.369).

Note: H-M4 ran in simulation mode because H-M3 did not produce trained checkpoints during ablation mode execution.

5.0.6. 5.6 SUMMARY

Causal chain verification:

- Steps 1-2 (mechanistic differences): VERIFIED
- Step 3 (*landscape* $\rightarrow$ *attractors*): NOT VERIFIED
- Step 4 (*attractors* $\rightarrow$ *profiles*): PARTIALLY VERIFIED

6. Discussion

6.0.1. 6.1 THE GAP BETWEEN MECHANISM AND MANIFESTATION

The mechanistic differences between RLHF and DPO were verified (H-M1, H-M2), but these did not produce distinct behavioral signatures (H-M3, H-M4). Several factors may explain this gap:

Seed variance dominance: At 7B scale with the evaluated training configuration, random seed variation appears to dominate over training method variation. Models trained with the same method but different seeds were less similar to each other than models trained with different methods.

LoRA constraints: The low-rank adaptation may constrain the parameter space sufficiently that both methods converge to similar regions despite different optimization paths. Full fine-tuning might permit larger divergence.

Benchmark granularity: Standard benchmarks may aggregate over dimensions in ways that mask method-specific differences. Item-level analysis or custom probes might reveal finer-grained distinctions.

Convergence dominance: The shared optimization objective (aligning with human preferences in HH-RLHF) may dominate over the different paths taken to reach it. Both methods ultimately optimize for the same preference signal.

6.0.2. 6.2 IMPLICATIONS FOR METHOD SELECTION

At 7B scale with LoRA fine-tuning, the choice between RLHF and DPO may matter less for final model behavior than previously assumed. The mechanistic differences are real but appear to wash out by evaluation time. Practitioners may reasonably select based on computational convenience (DPO avoids training a separate reward model) rather than expected behavioral differences.

Metric	DPO	RLHF	Threshold	Result
Margin std	0.343	0.208	-	DPO higher
Sharpness ratio	1.65	1.0	> 1.0	PASS
Boundary accuracy	0.589	-	> 0.55	PASS
Confident ratio	0.772	-	> 0.3	PASS

Metric	Value	Threshold	Result
Within-method mean similarity	0.967	-	-
Cross-method mean similarity	0.983	-	-
Clustering gap	-0.016	> 0.05	FAIL
Silhouette score	-0.411	> 0.1	FAIL
Cohen’s d	-1.295	d	> 0.3 PASS (opposite direction)
p-value	0.664	< 0.05	FAIL

6.0.3. 6.3 EVALUATION SENSITIVITY

The high profile correlation (0.978) between DPO and RLHF on standard benchmarks suggests these benchmarks may not be sensitive to method-specific alignment differences. More granular evaluation approaches—item-level analysis, specialized probes, or custom metrics targeting known mechanistic differences—may be needed to detect such differences if they exist.

6.0.4. 6.4 LIMITATIONS

Quick validation mode: H-M3 and H-M4 used abbreviated validation due to compute constraints. Results require replication with full multi-seed GPU training.

Scale constraint: All experiments used Llama-2-7B with LoRA. Larger models (13B, 70B) or full fine-tuning may show different patterns.

Single dataset: Only Anthropic HH-RLHF was used. Dataset characteristics may influence the comparative behavior of the methods.

Single-epoch training: Longer training might amplify method differences that are subtle after one epoch.

Simulation mode: H-M4 ran on simulated data because H-M3 did not produce trained checkpoints during execution. The benchmark comparison results require replication with actual trained models.

7. Conclusion

This study verified that RLHF and DPO exhibit genuinely different training dynamics: RLHF produces smooth, continuous reward predictions while DPO preserves sharper preference boundaries. However, at 7B scale with LoRA fine-tuning, these mechanistic differences did not manifest as distinct alignment signatures on standard evaluation benchmarks.

This pattern—”different dynamics, similar destinations”—suggests that the choice between RLHF and DPO may be less consequential for final model behavior than the mechanistic differences would suggest, at least under the tested conditions. The conditions under which preference learning mechanisms matter for final model behavior remain an open question for future investigation, potentially requiring larger scale, full fine-tuning, extended training, or more sensitive evaluation methodology.

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Benchmark	DPO Accuracy	RLHF Accuracy	Cohen’s d	p-value
TruthfulQA	0.362	0.320	+0.089	< 0.001
HH-helpful	0.612	0.704	-0.194	< 0.001
HH-harmless	0.623	0.633	-0.021	0.140

Hypothesis	Gate Type	Result	Pass Rate
H-E1: Benchmark Independence	MUST_WORK	PASSED	100%
H-M1: RLHF Smoothing	MUST_WORK	PASSED	100%
H-M2: DPO Sharpness	SHOULD_WORK	PASSED	100%
H-M3: Different Attractors	SHOULD_WORK	PARTIAL	25%
H-M4: Differential Profiles	SHOULD_WORK	PARTIAL	50%

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